



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
REVIEW TEST
10-07-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TOAS]

Time : 3 Hours

PAPER-2

Maximum Marks : 198

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 18)

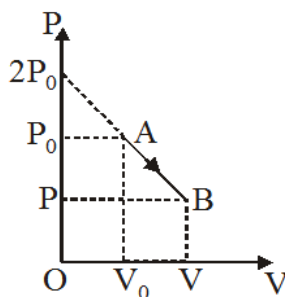
- This section contains **SIX (06)** questions.
- Each question has Four options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. One mole of an ideal gas is expanded from the state A (P_0, V_0) to final state B having volume V . The process follows a path represented by a straight line on the P-V diagram as shown. Up to what volume (V) the gas shall be expanded so that final internal energy is half the maximum internal energy during the process.

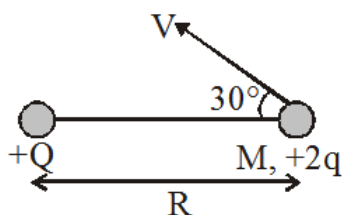


- (A) $\left(\frac{\sqrt{2}+1}{\sqrt{2}}\right) V_0$ (B) $\sqrt{2}V_0$
- (C) $(\sqrt{2}+1) V_0$ (D) $\left(\frac{\sqrt{2}+1}{2}\right) V_0$
2. A straight segment OP of length L of a circuit carrying current I ampere is placed along x-axis. Two infinitely long straight wires A and B each extending from $z = -\infty$ to $+\infty$ are fixed at $y = -\ell$ meter and $y = +\ell$ meter respectively. Wires A and B, each carry current I ampere along positive z-axis. Given that O is origin of the coordinate system. The magnitude of force on segment OP is :-

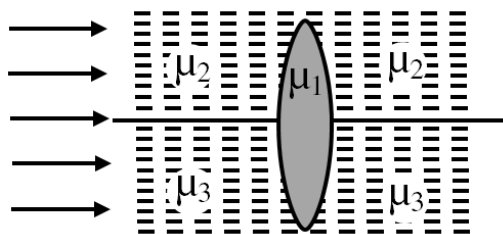
(A) $\frac{\mu_0}{\pi} I^2 \ell \ln \left(1 + \frac{L^2}{\ell^2}\right)$ (B) $\frac{\mu_0}{2\pi} I^2 \ell \ln \left(1 + \frac{L^2}{\ell^2}\right)$ (C) $\frac{\mu_0}{\pi} I^2 \ell \ln \left(1 + \frac{\ell^2}{L^2}\right)$ (D) $\frac{\mu_0}{2\pi} I^2 \ell \ln \left(1 + \frac{\ell^2}{L^2}\right)$

Space for Rough Work

3. In the diagram shown, the charge $+Q$ is fixed. Another charge $+2q$, is projected from a distance R from the fixed charge. The minimum separation between the two charges when the velocity becomes $\frac{1}{\sqrt{3}}$ times of the projected velocity at that instant, is (Assume gravity to be absent)



- (A) $\frac{\sqrt{3}}{2}R$ (B) $\sqrt{3}R$ (C) $\frac{1}{2}R$ (D) $4R$
4. An equi-convex lens is made of a material of refractive index μ_1 and is placed inside two liquids of refractive indices μ_2 and μ_3 as shown with $\mu_2 > \mu_1 > \mu_3$. A wide, parallel beam of light is incident on the lens from the left. The lens will give rise to :

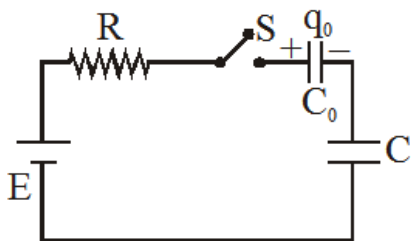


- (A) A single convergent beam. (B) Two different convergent beams.
(C) Two different divergent beams (D) A convergent and divergent beam.
5. A uniformly charged non-conducting rod is suspended vertically at its end. The rod can swing freely in the vertical plane without any friction. The linear charge density on the rod is $\lambda C/m$ and it has a uniformly distributed mass M . Length of the rod is L . A uniform horizontal field E is switched on so that the rod just manages to make itself horizontal. What is the maximum angular speed of the rod during its motion ?

- (A) $\sqrt{\frac{3(\sqrt{3}-1)g}{L}}$ (B) $\sqrt{\frac{3(\sqrt{2}-1)g}{2L}}$ (C) $\sqrt{\frac{3(\sqrt{2}-1)g}{L}}$ (D) $\sqrt{\frac{2(\sqrt{3}-1)g}{L}}$

Space for Rough Work

6. In the circuit shown, the switch S is open initially. The capacitor C is uncharged but C_0 has a charge q_0 at $t = 0$



- (A) The current immediately after switch S is closed is $I_C = \frac{\left(E + \frac{q_0}{C_0}\right)}{R}$.
- (B) The current immediately after switch S is closed is $I_c = \frac{E}{R}$
- (C) Final charge on C after closing switch is $\frac{EC_0C}{C_0 + C}$
- (D) Final charge on C after closing switch is $\frac{EC_0C}{C_0 + C} \left(1 - \frac{q_0}{C_0E}\right)$

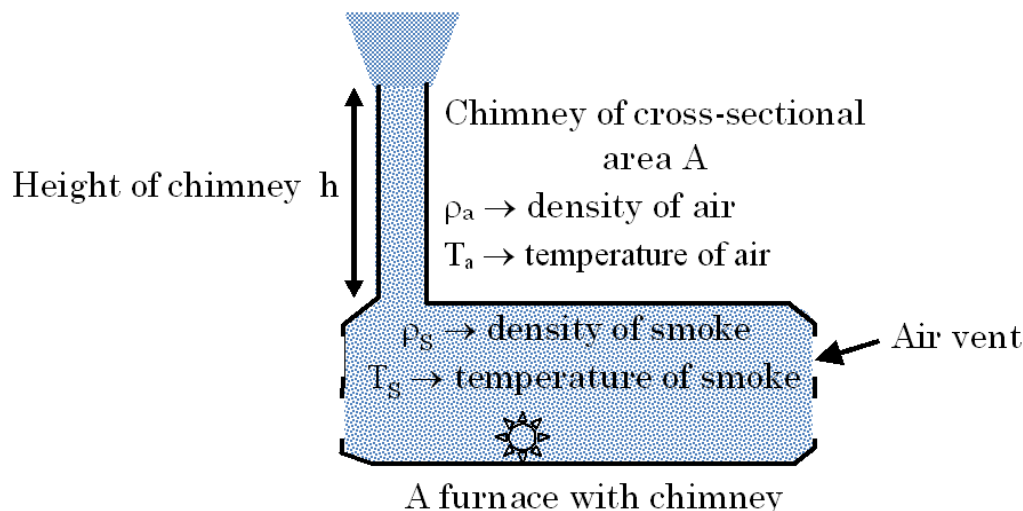
Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- This section contains **THREE** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has Four options (A), (B), (C) and (D) **ONE OR MORE THAN ONE** of these four option(s) is(are) correct.
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

Paragraph for Questions 7 and 8

Gaseous products of burning are released into the atmosphere of temperature T_a through a high chimney of cross section A and height h as shown in the figure. The solid matter is burnt in the furnace which is at temperature T_s . due to burning smoke (gas) generation rate (volume per unit time) is Q . It can be assume that



- $\rightarrow$ the velocity of the smoke (gas) in the furnace is negligibly small.
- $\rightarrow$ the density of the smoke (gas) does not differ from that air at the same temperature and pressure.
- $\rightarrow$ the gases can be treated as ideal while in furnace.
- $\rightarrow$ the pressure of the air changes with height according to the hydrostatic law.
- The change of the density of the air with height is negligible.
- $\rightarrow$ the flow of gases in the chimney follows Bernoulli's equation.
- $\rightarrow$ the change of the density of the gas (smoke) is negligible throughout the chimney.

7. What is the minimum height of the chimney needed in order that chimney functions efficiently, so that it can release all of the produced smoke (gas) into the atmosphere ? Express your answer in terms of Q , A , g , T_a , ρ_a , ρ_s and ΔT , where $\Delta T = T_s - T_a$

(A) $\frac{\rho_s T_a Q^2}{g A^2 \Delta T (\rho_a - \rho_s)}$ (B) $\frac{T_a Q^2}{2 g A^2 \Delta T}$ (C) $\frac{Q^2 \rho_s}{2 (\rho_a - \rho_s) A^2 g}$ (D) $\frac{Q^2 (T_a + \Delta T)}{2 g A^2 \Delta T}$

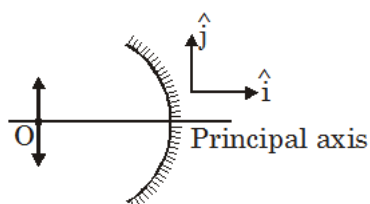
8. Assume that two chimneys are built to serve exactly the same purpose. Their cross sections are identical, but are designed to work in different parts of the world, one in cold regions designed to work at an average atmospheric temperature of -23°C and the other in warm regions, designed to work at an average atmospheric temperature of 27°C . The temperature of the furnace is 327°C . It was calculated that the height of the chimney designed to work in cold regions is 100 m. How high is the other chimney ?

(A) 71.4 m (B) 280 m (C) 140 m (D) 100 m

Space for Rough Work

Paragraph for Questions 9 and 10

A point object is placed at a distance 50 cm from the pole of a concave mirror. The radius of curvature of mirror is 30 cm. Point object O oscillates with amplitude of 1mm perpendicular to the principal axis as shown.



9. The amplitude of image is :-

- (A) $\frac{3}{7}$ mm (B) $\frac{2}{7}$ mm (C) $\frac{4}{3}$ mm (D) $\frac{11}{7}$ mm

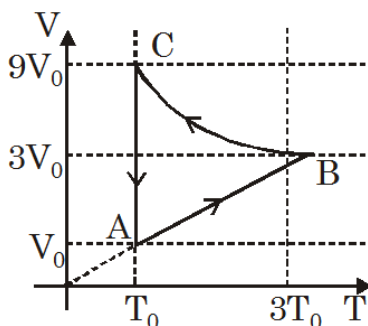
10. Phase difference between motion of object and image when object crosses the principal axis is :-

- (A) π (B) 0 (C) $\frac{\pi}{2}$ (D) None of these

Space for Rough Work

Paragraph for Questions 11 and 12

Two moles of an ideal mono-atomic gas is taken through a cyclic process ABCA as shown in the V-T diagram. In the process BC, $VT = \text{constant}$. The temperature at A is $T_0 = 300 \text{ K}$. [take $\ln(3) = 1.1$]. Then answer the following questions.



11. The work done by the gas during the process 'BC' is :-
 (A) $600 R$ (B) $900 R$ (C) $1200 R$ (D) $1500 R$
12. The efficiency (in percentage) of the cycle is :-
 (A) 12 (B) 18 (C) 24 (D) 36

Space for Rough Work

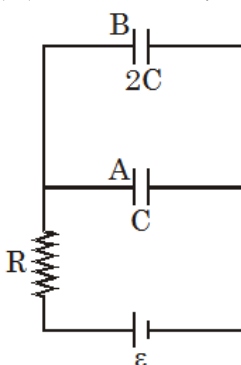
SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

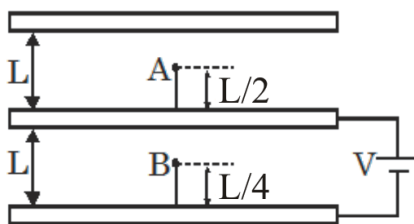
Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. In the circuit shown in the figure, find the heat generated (in μJ) through the resistor upto the instant when charge on capacitor A is half of the maximum values that the capacitors would store finally. (Initially both the capacitor are uncharged) (Given : $C = 8 \mu\text{F}$, $\varepsilon = 3 \text{ volt}$).

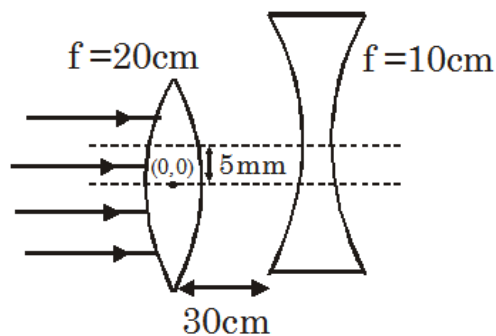


2. In a gravity free space, three identical metal plates have been placed parallel to each other as shown. In the initial state shown in the diagram, the potential difference between the points A and B is V_1 . Now, the middle plate is shifted down a distance $L/2$, keeping the battery connected. The other plates are kept fixed. The potential difference between the points A and B now becomes V_2 . The value $\frac{6V_1}{V_2}$ is equal to

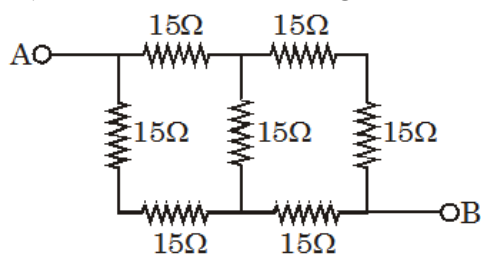


Space for Rough Work

3. If the optical axis of convex and concave lenses are separated by a distance 5 mm shown in the figure. If the coordinates of final image formed by the combination of Lenses is (x, y) when the Parallel Beam of Light is incident on Convex Lens. If origin is at the Optical centre of convex lens, find $x + 4y$ (in cm).

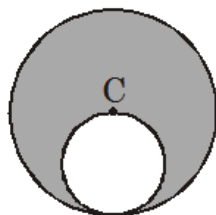


4. Find the equivalent resistance (in Ω) between A and B in the given network.



Space for Rough Work

5. A solid sphere of radius 'R' is uniformly charged with charge density ρ in its volume. A spherical cavity of radius $\frac{R}{2}$ is made in the sphere as shown in the figure. Electric potential at the centre of the sphere is $\frac{A\rho R^2}{B\epsilon_0}$. Where A and B are positive integers. Find A^2B .



6. Two particles A and B of mass 1kg and 2kg respectively are kept 1 meter apart and are released to move under mutual attraction. When speed of B is 3.6 cm/hour, what is the separation (in cm) between the particles. Take $G = \frac{20}{3} \times 10^{-11} \text{ Nm}^2/\text{kg}$.

Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 18)

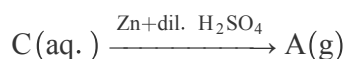
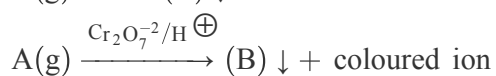
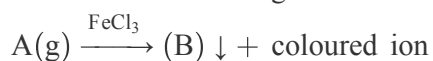
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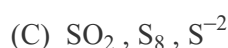
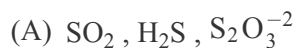
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. Consider the following reactions :-



Identify A, B & C respectively.



2. Consider the following reaction $2A(g) \rightarrow 3B(g) + C(g)$. Starting with pure A having pressure 2 atm initially, the total pressure is exactly doubled in 2 hrs. The possible order of reaction is :-

(A) zero

(B) first

(C) second

(D) third

3. The compressibility factor of a Vanderwaal gas is 0.5 at $27^\circ C$ and 24 atm. The value of Vanderwaal constant 'a' (in $\text{atm-L}^2/\text{mol}^2$) is
(b = 0.10 L/mol, R = 0.08 L-atm/K-mol)

(A) 2.4

(B) 4.0

(C) 3.6

(D) 9.0

Space for Rough Work

4.

	Conversion	Reagent used for conversion
(a)	$\text{Me}-\overset{\text{O}}{\underset{\text{O}}{\text{C}}}-(\text{CH}_2)_2-\text{CO}_2\text{Et} \longrightarrow \text{Me}-\underset{\text{OH}}{\text{CH}}-(\text{CH}_2)_2-\text{CO}_2\text{Et}$	R ₁
(b)	$\text{O}_2\text{N}-(\text{CH}_2)_2-\text{CH}=\text{CH}_2 \longrightarrow \text{H}_2\text{N}-(\text{CH}_2)_2-\text{CH}=\text{CH}_2$	R ₂
(c)	$\text{O}_2\text{N}-(\text{CH}_2)_2-\text{CN} \longrightarrow \text{O}_2\text{N}-(\text{CH}_2)_2-\text{CH}_2-\text{NH}_2$	R ₃

 R₁, R₂ & R₃ are respectively :

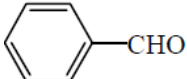
 (A) LiAlH₄, NaBH₄, B₂H₆

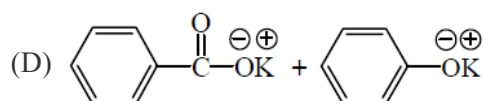
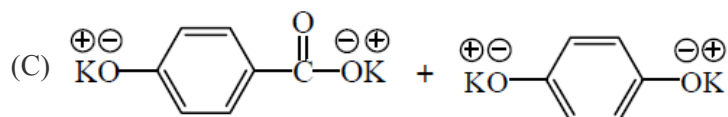
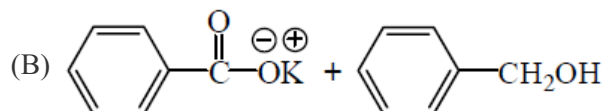
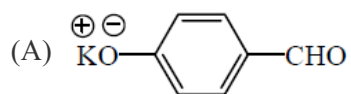
 (B) B₂H₆, NaBH₄, LiAlH₄

 (C) NaBH₄, LiAlH₄, B₂H₆

 (D) LiAlH₄, B₂H₆, NaBH₄

Space for Rough Work

5. Which products are formed when compound  is treated with concentrated aqueous KOH solution?



6. Assuming that during the formation of $XY^{\oplus}$ from XY the bond order increase where as during the formation of $Z_2^{\oplus}$ from Z_2 the bond order decrease.
Based on above information which of the following statement is incorrect?

- (A) During the formation of XY^+ from XY the electron is lost from antibonding molecular orbital
(B) During the formation of $Z_2^{\oplus}$ from Z_2 the electron is lost from bonding molecular orbital
(C) In above change XY undergoes in oxidation
(D) In above change Z_2 undergoes in reduction

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- This section contains **THREE** paragraphs.
- Based on each paragraph, there are **TWO** questions
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- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

Paragraph for Questions 7 and 8

The crystal field splitting, Δ_o depends upon the field produced by the ligand and charge on the metal ion. Some ligands are able to produce strong fields in which case, the splitting will be large whereas others produce weak field and consequently result in small splitting of d-orbitals.

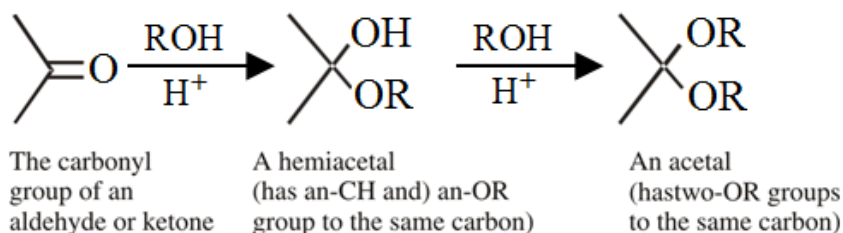
The influence of the ligand on the colour of a complex may be, illustrated by considering the $[\text{Ni}(\text{H}_2\text{O})_6]^{+2}$ complex. (which forms when nickel (II) chloride is dissolved in water) and reaction of it with ethane 1, 2-diamine (en).

7. If the bidentate ligand ethane 1,2-diamine (en) is progressively added in green aqueous solution of $[\text{Ni}(\text{H}_2\text{O})_6]\text{Cl}_2$ in the molar ratios en : Ni as, 1 : 1 , 2 : 1 , 3 : 1 then their associated colour are respectively :
- | | |
|------------------------------|-------------------------------|
| (A) Pale blue , Violet, Blue | (B) Violet , Pale blue , Blue |
| (C) Violet , Blue, Pale blue | (D) Pale Blue, Blue, Violet |
8. Sum of the stereo isomeric forms possible for complexes which are formed in above question :
- | | | | |
|-------|-------|-------|-------|
| (A) 5 | (B) 2 | (C) 4 | (D) 3 |
|-------|-------|-------|-------|

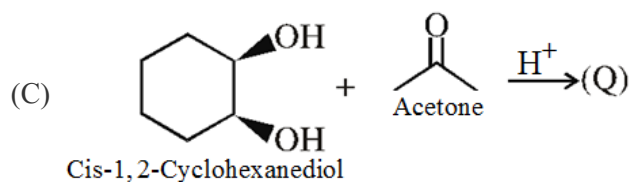
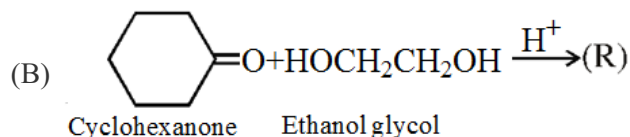
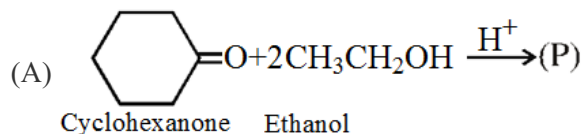
Space for Rough Work

Paragraph for Questions 9 and 10

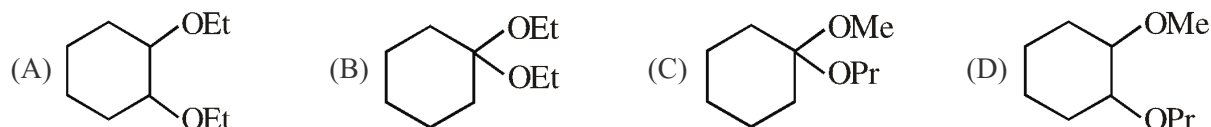
Aldehydes and ketones react with an alcohol from compounds called hemiacetals, in which there is one hydroxyl group and one ether-like group. Reaction of a hemiacetal with a second molecule of alcohol gives an acetal and a molecule of water. We study this reaction



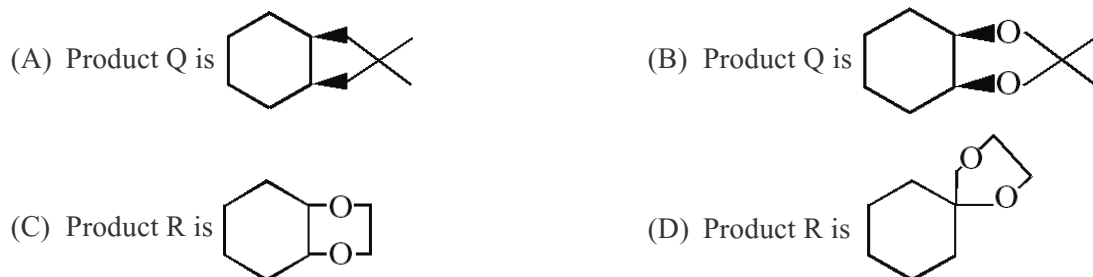
Draw structural formulas for the hemiacetal and acetal formed from these reagents. The stoichiometry of each reaction is given in the problem.



9. Product P is not possible -



10. Which of the following statement is/are correct?



Space for Rough Work

Paragraph for Questions 11 and 12

Enthalpy of combustion of a given compounds is the change in enthalpy when 1 mole of this compound combines with required amount of O_2 to give product in their stable forms. The enthalpy of combustion is usually measured by placing a known mass of the compound in a closed steel container (known as bomb-calorimeter). Graphite is thermodynamically most stable form of carbon.

11. Given the data below :

$$\Delta H_{\text{comb}}^0(\text{CH}_4, \text{g}) = -900 \text{ kJ/mole}$$

$$\Delta H_{\text{comb}}^0(\text{C, graphite}) = -400 \text{ kJ/mole}$$

$$\Delta H_{\text{comb}}^0(\text{H}_2, \text{g}) = -300 \text{ kJ/mole}$$

$$\Delta H_{\text{atomization}}^0(\text{C, graphite}) = 700 \text{ kJ/mole}$$

$$\text{and } \Delta H_f^0(\text{H, g}) = 220 \text{ kJ/mole}$$

Identify the incorrect option(s) :

(A) Bond enthalpy of C–H bond is 340 kJ/mole

(B) 1 mole graphite produces more heat on combustion than 1 mole diamond at 1 atm and 25°C

(C) Enthalpy of atomisation of $\text{CH}_4(\text{g})$ is 1600 kJ/mole

(D) Bond enthalpy of C–H bond is 420 kJ/mole

12. Identify the correct statement -

(A) $\Delta H_{\text{comb}}^0(\text{C, graphite})$ is equal to $\Delta H_f^0(\text{CO}_2, \text{g})$

(B) $\Delta H_f^0(\text{H}_2\text{O}, \ell)$ is equal to $\Delta H_{\text{comb}}^0(\text{H}_2, \text{g})$

(C) The standard enthalpy of combustion is always negative.

(D) The standard enthalpy of formation is always negative

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SECTION-II : (Maximum Marks: 24)

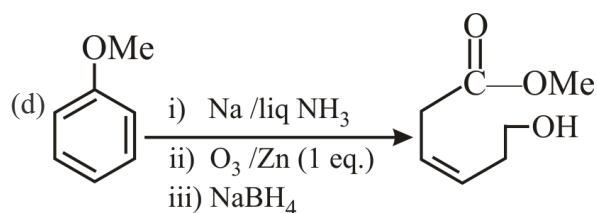
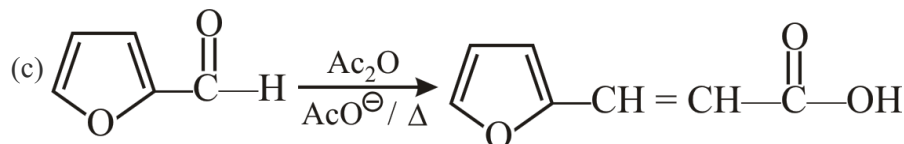
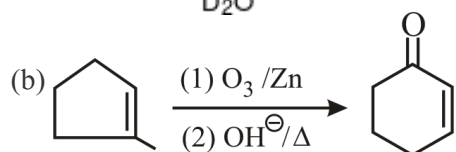
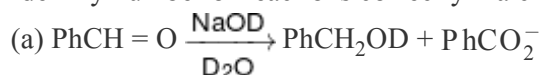
- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

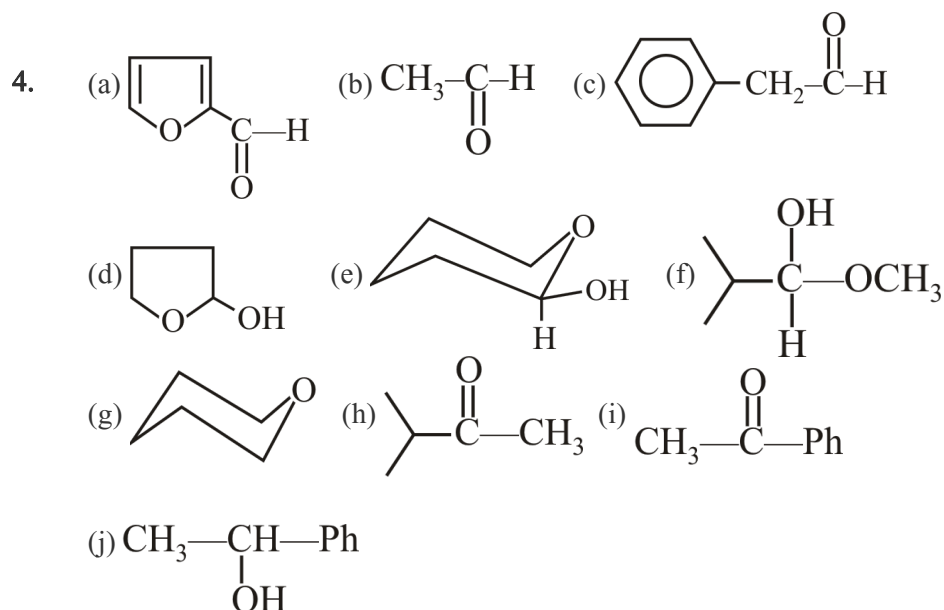
Zero Marks : 0 In all other cases.

1. A solution containing 30 g of a non volatile solute in exactly 90 g water has a vapour pressure of 20 mm Hg at 25°C. Further 18 g of water is then added to the solution. The resulting solution has vapour pressure of 21 mm Hg at 25 °C. Calculate molar mass of the solute.
2. If zinc blende (ZnS) crystal is heated, all ions present on any one face are lost. Monovalent anion (Q^-) enters each unit cell to maintain electrical neutrality without disturbance. If new formula of unit cell becomes $Zn_xS_yQ_z$, then $(x + y + z)$ is :-

3. Identify number of reactions correctly match with their products ?



Space for Rough Work



Answer in two digit form XY

X = Number of compounds giving (+) test with $\text{AgNO}_3 / \text{NH}_4\text{OH}$

Y = Number of compounds giving yellow colour with $\text{I}_2 + \text{NaOH}$

5. Consider the following compounds.

$\text{Pb}(\text{NO}_3)_2$, LiNO_3 , NaNO_3 , NH_4NO_2 , KMnO_4 , $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$, HNO_3 (conc.), $(\text{NH}_4)_2\text{CO}_3$

If total number of compound(s) which produce coloured gas on heating = x

If total number of compound(s) which produce atleast one paramagnetic gaseous product = y

then find $\frac{x}{y} = ?$

6. Consider the following anions?

CO_3^{2-} , SO_3^{2-} , S^{2-} , $\text{S}_2\text{O}_3^{2-}$, NO_2^- , HCO_3^- , HSO_3^- , CH_3COO^-

If total number of anion which produce colourless solution with KMnO_4/H^+ by redox reaction = A

If total number of anion which produce coloured volatile product with dil. HCl = B

Then find $\frac{A+B}{2} = ?$

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- Each question has Four options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. If $\frac{\cos^4 x}{\theta_1} + \frac{\sin^4 x}{\theta_2} = \frac{1}{(\theta_1 + \theta_2)}$, then $\frac{d\theta_2}{d\theta_1}$ equals (where x is not dependent on θ_1 or θ_2)
 (A) $\sin^2 x$ (B) $\cos^2 x$ (C) $\tan^2 x$ (D) $\sec^2 x$
2. Let $F(x) = C \sin x \int_0^x \cos t dt + 2 \int_0^x t dt + C \cos^2 x - x^2$ (where $C \in \mathbb{R}^+$). If $x^2 - 2x + 3 \geq F(x) \forall x \in \mathbb{R}$, then the maximum value of the function $\left(\frac{6e^x - e^{2x} - 1}{e^x} \right)^{F(2)}$ is equal to -
 (A) 2 (B) 4 (C) 16 (D) 32
3. Suppose that f is differentiable function such that $f'(x) \leq 2$ for all x. If $f(1) = 2$ and $f(4) = 8$ then $f(2)$ has the value equal to
 (A) 4 (B) 2 (C) 1 (D) 0

Space for Rough Work

4. Let $f(x)$ be a differentiable function in $[-1, \infty)$ and $f(0) = 1$ such that $\lim_{t \rightarrow x+1} \frac{t^2 f(x+1) - (x+1)^2 f(t)}{f(t) - f(x+1)} = 1$, then the value of $\lim_{x \rightarrow 1} \frac{\ln(f(x)) - \ln 2}{x - 1}$ is
 (A) 0 (B) 1 (C) 2 (D) does not exist
5. If $f(x) = (\sin x)^{1/3} \cdot (\sin 2x)^{1/2} \cdot \sin 3x$, then value of $f' \left(\frac{\pi}{3} \right)$ is of the form $-\frac{3^a}{2^b}$, then value of $\left(a - \frac{b}{2} \right)$ is
 (A) 1 (B) 2 (C) 5 (D) 0
6. Let $\int \frac{x^2 + 20}{(x \sin x + 5 \cos x)^2} dx = f(x) + \tan x + C$, $f(0) = 0$ where C denotes constant of integration. Then the value of $\pi + 5f(\pi)$ is equal to
 (A) 2 (B) 4 (C) 3 (D) 0

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- This section contains **THREE** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has Four options (A), (B), (C) and (D) **ONE OR MORE THAN ONE** of these four option(s) is(are) correct.
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

Paragraph for Questions 7 and 8

Let $f(x) = 3x^2$ and $g(x) = 2x^3 + 1$ are two curves and L is the equation of tangent at the point where $f(x)$ and $g(x)$ touch each other

7. Which option(s) is/are correct -
 (A) Equation of L is $6x - y = 3$
 (B) Equation of L is $x + 2y = 7$
 (C) Equation of normal to the curve $g(x)$ from (7, 2) is $x + 6y = 19$
 (D) Equation of normal to the curve $f(x)$ from (7, 2) is $x + 6y = 19$
8. Which option(s) is/are correct -
 (A) Angle between $f(x)$ and $g(x)$ at the point of intersection is $\tan^{-1} \frac{9}{7}$
 (B) Angle between $f(x)$ and $g(x)$ at the point of intersection is $\sin^{-1} \frac{9}{\sqrt{130}}$
 (C) Point of intersection of L and $g(x)$ is $(-2, -15)$
 (D) Value of $\lim_{x \rightarrow 1} \frac{g(x) - f(x)}{(x - 1)^2}$ is 3

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Paragraph for Questions 9 and 10

Let $f(x) = ax^3 + bx^2 + cx + d$, satisfying $\lim_{x \rightarrow 0} \left(3 + \frac{f(x)}{x^2} \right)^{\frac{1}{x}} = e^2$, then

On the basis of above information, answer the following questions :

9. Correct option's is /are -
- (A) $a + b = 0$ (B) $a + b + c + d = 0$
 (C) $a + c = 2$ (D) $d - b = 2$
10. Sum of all roots of the all possible equations $f(x) - 2(r^2 - 1)x^2 = 0$ (where $r = 1, 2, \dots, 10$)
- (A) 385 (B) 250 (C) 165 (D) 305

Paragraph for Questions 11 and 12

Consider $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function such that $f(x) = x + \int_0^x e^{-t} f(x-t) dt$.

11. Minima of function $y = f(x)$ occurs at
- (A) $x = -1$ (B) $x = \frac{1}{2}$ (C) $x = 1$ (D) $x = 2$
12. The value of $\int_1^2 \frac{f''(x)}{x^2} dx$ is less than
- (A) 1 (B) $\frac{1}{4}$ (C) $-\frac{1}{2}$ (D) $\frac{3}{4}$

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SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. If $\int \frac{e^x (x^2 - x + 1)}{(x^2 + 1) \sqrt{x^2 + 1}} dx = e^x \cdot f(x) + C$, where C is constant of integration, then number of integers in the range of $f(x)$ is
2. If $n \in \mathbb{N}$ and $f(x) = ({}^nC_0x + 1) ({}^nC_1x + 2) ({}^nC_2x + 3) ({}^nC_3x + 4) \dots ({}^nC_nx + n + 1)$ and $f'(0) = \frac{3}{2}(7!)$, then n is equal to
3. Let $P(x)$ be a polynomial of degree 4 having a local maximum at $x = 2$ and $\lim_{x \rightarrow 0} \left(3 - \frac{P(x)}{x} \right) = 27$. If $P(1) = -9$ and $P''(x)$ has a local minimum at $x = 2$. If global maximum value of $y = P'(x)$ on the set $A = \{x : x^2 + 12 \leq 7x\}$ is $4M$ then value of M is :

Space for Rough Work

4. If $0 < x < 1$, then the solution of equation $3\sin^{-1}\left(\frac{2x}{1+x^2}\right) + 2\tan^{-1}\left(\frac{2x}{1-x^2}\right) - 4\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) = \frac{\pi}{4}$ is
5. For $f(x) = x^4 + |x|$, Let $I_1 = \int_0^{\pi} f(\cos x)dx$, $I_2 = \int_0^{\pi/2} f(\sin x)dx$, then the value of $\frac{I_1}{I_2}$ is
6. If $f(x) = \begin{cases} x^2 - 5x + 6 & , \quad 2 < x < 3 \\ px^2 + qx + 1 & , \quad 3 \leq x \end{cases}$ is differentiable at $x = 3$, then value of $(p - q)$ is
-

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them._____
Signature of the CandidateI have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same._____
Signature of the Invigilator

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